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Thermodynamic modeling of the C–RE (RE = La, Ce and Pr) systems

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ABSTRACT

The phase diagrams and thermodynamic properties of the C–RE (RE = La, Ce and Pr) binary systems were assessed by using the CALPHAD method applied to the experimental data in the literature. In the C–RE (RE = La, Pr) systems, RE_2C_3 and βREC_2 (high-temperature form of REC_2 phase) were modeled as non-stoichiometric phases while αREC_2 (low-temperature form of REC_2 phase) was considered to be stoichiometric phase. For La_2C_3 phase, enthalpy increment (H_T–H_{298K}) information was also considered in the optimization. Due to their negligible homogeneities, intermediate compounds Ce_2C_3 and CeC_2 in the C–Ce system, were treated as stoichiometric phases. One set of self-consistent thermodynamic parameters was finally obtained for each of these binary systems. Comparisons between calculated and measured phase diagrams as well as thermodynamic properties showed that most of the experimental information can be satisfactorily accounted for by the present thermodynamic descriptions.

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1. Introduction

Rare earth (RE) elements are important modifiers of steels and ceramics. Rare earth metals can take deep deoxidation, deep desulfurization and remove harmful elements P, S, As, Sb, Bi, Pb, Sn or depress their nocuous actions in steels [1]. With a small amount of additional RE elements, toughness, wear resistance and corrosion resistance of steels or ceramics can be significantly improved [1,2]. Therefore, knowledge about phase equilibria and thermodynamic properties of the carbon–RE systems is very important.

Up to now, several groups have, more or less, studied parts of equilibrium phase diagrams of the C–La [3], C–Ce [4] and C–Pr [5] systems. However, in order to gain a full understanding of these systems, it is necessary to have thermodynamic descriptions by means of the CALPHAD method [6]. Consequently, the present work is devoted to obtaining self-consistent thermodynamic parameters for the binary C–La, C–Ce and C–Pr systems by means of the CALPHAD approach.

2. Evaluation of experimental information

2.1. The C–La system

By means of thermal analysis, metallographic, X-ray diffraction (XRD), dilatometric, and electrical resistance measurements, Spedding et al. [3] examined the entire C–La system and proposed a phase diagram containing two intermetallic compounds, La_2C_3 and LaC_2. The crystal structure data [7–11] about all the phases in the C–La system are listed in Table 1. The solubility limits of C in (La) at 695 and 775 °C were approximately determined by examining the photomicrographs of low-carbon alloys via a quantitative microscopic point-count method [3]. By plotting lattice parameters versus C concentration of the furnace cooled alloys around the La_2C_3 composition, Spedding et al. [3] determined the homogeneity range of La_2C_3 at room temperature. As for LaC_2, Spedding et al. [3] found that its lattice parameters vary slightly with composition at room temperature by using the same method as that of La_2C_3 phase. Through carbon analysis, thermal analysis and X-ray technique of quenched alloys, Spedding et al. [3] observed that both La_2C_3 and LaC_2 phases exhibit certain homogeneity range at higher temperatures.

Spedding et al. [3] measured the lower transition temperature in lanthanum ($(\alpha La) \leftrightarrow (\beta La)$) by dilatometric measurements and found the value to be 310 ± 5 °C. Their data on cooling was quite low due to the overcooling. Consequently, they only reported the transition temperature on heating. Recently, a lower value of 277 °C from SGTE database, reported by Dinsdale [20], is now widely accepted and adopted in the present work. Two eutectic reactions, $L \rightarrow (\beta La) + La_2C_3$ and $L \rightarrow \beta LaC_2 + (C)$, were found in the C–La system by Spedding et al. [3] with the reaction temperatures of 806 ± 2 °C, determined by drawing cooling and heating curves and 2271 ± 15 °C using the Pirani and Alterthums method, respectively. By means of metallographic and thermal

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Table 1
Crystal structure data for the solid phases in the C–RE (RE = La, Ce, Pr) systems.

System	Phase	Pearson symbol	Space group	Prototype	Lattice parameter (Å)			Reference
					a	b	c	
C–La	(α La)	*hP*4	*P*63/*mmc*	α La	3.77	–	12.13	[7]
	(β La)	*cF*4	*Fm*-3*m*	Cu	5.29	–	–	[8]
	(γ La)	*cI*2	*Im*-3*m*	W	4.26	–	–	[7]
	La_2C_3	*cI*40	*I*4-3*D*	Pu_2C_3	8.81	–	–	[9]
	α LaC_2	*tI*6	*I*4/*mmm*	CaC_2	3.93	–	6.58	[10]
	β LaC_2	*cF*12	*Fm*-3*m*	CaF_2	6.01	–	4.98	[11]
C–Ce	(γ Ce)	*cF*4	*Fm*-3*m*	Cu	5.15	–	–	[12]
	(δ Ce)	*cI*2	*Im*-3*m*	W	4.12	–	–	[7]
	Ce_2C_3	*cI*40	*I*4-3*D*	Pu_2C_3	8.44	–	–	[13]
	α CeC_2	*tI*6	*I*4/*mmm*	CaC_2	3.88	–	6.48	[14]
	β CeC_2	*cF*12	*Fm*-3*m*	CaF_2	5.93	–	5.20	[15]
C–Pr	(α Pr)	*hP*4	*P*63/*mmc*	α La	3.67	–	11.83	[16]
	(β Pr)	*cI*2	*Im*-3*m*	W	4.13	–	–	[7]
	Pr_2C_3	*cI*40	*I*4-3*D*	Pu_2C_3	8.59	–	–	[17]
	α PrC_2	*tI*6	*I*4/*mmm*	CaC_2	3.85	–	6.42	[18]
	β PrC_2	…	…	…	…	…	…	…
	(C)	*hP*4	*P*63/*mmc*	C (graphite)	2.52	–	4.12	[19]

Table 2
Experimentally measured values of enthalpies of formation at 298 K for the intermetallic compounds in the C–RE (RE = La, Ce, Pr) systems.

Compound	$\Delta_f H$ (298 K), kJ/mol-atom	Method	Reference
La_2C_3	−38.6	emf	[40]
LaC_2	−19.5 ± 3.4	CO pressure	[32]
	−26.5	Knudsen effusion	[33]
	−27.2 ± 1.4	Calorimetry	[34]
	−29.1 ± 3.5	emf	[35]
	−29.7 ± 8	Knudsen effusion	[36]
	−50.3 ± 5.3	Langmuir evaporation	[37]
Ce_2C_3	−35.3 ± 1.1	Bomb calorimetry	[41]
CeC_2	−20.9 ± 7	Knudsen cell MS	[42]
	−27.2	Knudsen cell MS	[15]
	−25.4 ± 1.4	Synthesis calorimetry	[43]
	−29.5 ± 3.5	emf	[35]
	−31.3	Controlled CO pressure	[44]
	−32.4 ± 1.8	Bomb calorimetry	[41]
	−34.9	Knudsen effusion	[33]
	−40.03 ± 3.7	Dynamic effusion MS	[45]
Pr_2C_3	−27.9	emf	[40]
PrC_2	−28.2 ± 3.5	emf	[35]
	−29.4 ± 1.6	Synthesis calorimetry	[43]

analysis methods, the eutectic composition of $L \rightarrow (\beta La) + La_2C_3$ was measured to be 20.6 at.% C [3]. For $L \rightarrow \beta LaC_2 + (C)$, a similar reaction temperature, 2230 ± 20 °C, was reported by Krikorian et al. [21] using thermal analysis technique. In addition, Spedding et al. [3] determined the temperature of peritectic reaction ($L + \beta LaC_2 \rightarrow La_2C_3$) to be 1415 ± 3 °C, which is in good agreement with 1430 ± 20 °C from Paderno et al. [22] and Yupko et al. [23]. By extrapolating the observed melting points of alloys near the theoretical composition of LaC_2, Spedding et al. [3] determined its melting point to be 2356 ± 25 °C. Later, Kosolapova and Makarenko [24] reported a higher value of 2438 ± 100 °C with a high uncertainty. Also, the liquidus up to the temperature at about 2050 °C was measured by Spedding et al. [3] using equilibrium-diffusion method.

With appropriate methods and smaller experimental errors, the phase diagram data over the whole composition range reported by Spedding et al. [3] are considered to be reliable. We thus adopted their work in the present modeling with one exception which is the polymorphic transformation temperature for LaC_2. It was estimated as 1800 ± 100 °C by Spedding et al. [3], but much lower values were reported from recently investigations [21,25–30]. By using the thermal analysis technique, Krikorian et al. [21,25] and Loe et al. [29] determined the polymorphic transformation temperature for LaC_2 to be 1060 ± 20 °C and 1075 ± 15 °C, respectively. A similar value of 1081 ± 2 °C was reported by Adachi et al. [27,28,30] using DTA technique. A even lower value of 995 °C was reported by McColm et al. [26] based only on their cooling curves. This low value is excluded in the present optimization because of its overcooling factor. Based on the above mentioned fact, 1081 ± 2 °C is selected in the present optimization due to its smallest experimental error resulting from accurate DTA measurement.

The enthalpy increments ($H_T - H_{298\,K}$) of La_2C_3 in the temperature range of 169–1471 °C were measured by Kriklya et al. [31] using a high-temperature differential calorimeter. In addition, the enthalpy of formation at 298.15 K for LaC_2 was measured by [32–37] and predicted by [38,39], while that for La_2C_3 was measured by Anderson and Bagshaw [40] using electron motive force (emf) method. The experimental methods and corresponding values of enthalpies of formation at 298.15 K for La_2C_3 and LaC_2 are listed in Table 2. Considering that the values of enthalpy of formation for LaC_2 are very scattering, only four values measured by Faircloth et al. [33], Meschel and Kleppa [34], Anderson and Bagshaw [35], Stearns and Kohl [36] are considered in the present optimization.

2.2. The C–Ce system

Based on only a few analysis data points and metallographic observations, Stecher et al. [4] roughly established the C–Ce

phase diagram by resembling it to the C–La system. Later, Anderson et al. [46] experimentally investigated this binary system. Only two intermetallic compounds, Ce_2C_3 and CeC_2, both with negligible homogeneities, were found by Anderson et al. [46]. The crystal structure data [7,12–15] about all the phases in the C–Ce system are listed in Table 1.

Stecher et al. [4] indicated the peritectic decomposition for Ce_2C_3 but providing no data for temperature. Paderno et al. [47] and Yupko et al. [23] reported a decomposition temperature of 1530 ± 30 °C but failed to specify whether it was congruent or incongruent melting. Subsequently, Kosolapova et al. [48] reported a slightly different value without mentioning the melting type. In the present work, the decomposition reaction for Ce_2C_3 is treated as a peritectic reaction at 1530 °C with uncertainty of 30 °C. By using a mass spectrometer-Knudsen cell technique combined with thermal analysis and XRD, Winchell and Baldwin [15] determined the polymorphic transformation temperature of CeC_2 and the eutectic reaction ($L \rightarrow \beta CeC_2$+(C)) temperature to be 1100±20 °C and 2270 ± 20 °C, respectively. These are slightly higher than the results (1090 ±20 °C and 2245 ± 20 °C) reported by Krikorian et al. [21,25] using thermal analysis method. In addition, similar values for the CeC_2 polymorphic transformation temperature were reported by other investigators, including 1090 °C by McColm et al. [26,29] and 1107 °C by Adachi et al. [30]. Because the experimental methods employed by Winchell and Baldwin [15] were more comprehensive and reliable, their experimental results are adopted in the present work. Kosolapova and Makarenko [24] reported a melting temperature of CeC_2 to be 2540 ± 100 °C. Later, this temperature was updated to be 2290 °C based on new experiments by Makarenko et al. [23,48–50]. This new result is utilized in the present work.

In addition, the enthalpy of formation at 298.15 K for CeC_2 was measured by [33,35,15,44,42,41,43,45] and predicted by [38], while that for Ce_2C_3 was measured by Baker et al. [41]. The experimental methods and corresponding values of enthalpies of formation at 298.15 K for Ce_2C_3 and CeC_2 are listed in Table 2. The values of enthalpy of formation for CeC_2 determined by Balducci et al. [42] and Vidhya et al. [45] are too far away from the other groups' results. In a consequence, these values are excluded from the present optimization. The enthalpies of formation for Ce_2C_3 and CeC_2 determined by one group of investigators [41], are considered to be reliable, and thus firstly utilized as the input experimental data in the present optimization.

2.3. The C–Pr system

The C–Pr phase diagram was firstly determined by Eremenko et al. [5] using differential thermal, XRD, and metallographic analysis. According to Eremenko et al. [5], the solubility limits of C in (αPr) and (βPr) are about 5 and 8 at.% C, respectively. Spedding et al. [51] found that Pr_2C_3 and βPrC_2 exhibit certain homogeneities while αPrC_2 exhibits negligible homogeneity based on the analysis of their lattice parameters. The crystal structure data [7,16–19] about all the phases in the C–Pr system are listed in Table 1.

Eremenko et al. [5] determined the temperature of peritectic reaction ($L + \beta PrC_2 \rightarrow Pr_2C_3$) to be 1545 ± 4 °C at about 50 at.% C, which is in good agreement with the reported temperature (1560 °C) by Kosolapova et al. [48]. In addition, two eutectic reactions were also found in the C–Pr system by Eremenko et al. [5], i.e. $L \rightarrow$ (βPr) + Pr_2C_3 and $L \rightarrow \beta PrC_2$ + (C) at 800 ± 4 °C at about 15 at.% C and 2254 ± 6 °C at about 71.5 at.% C, respectively. For $L \rightarrow \beta PrC_2$ + (C), a similar reaction temperature of 2255 ± 20 °C was reported by Krikorian et al. [21]. The melting temperature of PrC_2 was determined by Kosolapova and Makarenko [24] and Eremenko et al. [5] to be 2535 ± 100 °C and 2320 °C, respectively. Other results of the melting temperature of PrC_2 [49,50], ranging from 2147 to 2160 °C, about 100 °C lower than the PrC_2–(C) eutectic temperature [5,24] are considered to be unreliable in this work. Furthermore, Eremenko et al. [5] reported the polymorphic transformation temperature of PrC_2 to be 1145 ± 4 °C in the Pr_2C_3–PrC_2 region and 1134 ± 4 °C in the PrC_2–(C) region. Considering the fact that major experimental data in the C–Pr system reported by Eremenko et al. [5] agreed well with those of [3,21,48,52], these data [5] are utilized in the present optimization.

In addition, the enthalpy of formation at 298.15 K for Pr_2C_3 was measured by Anderson and Bagshaw [40], while that for PrC_2 by Anderson and Bagshaw [35] and Meschel and Kleppa [43]. The experimental methods and corresponding values of enthalpies of formation at 298.15 K for Pr_2C_3 and PrC_2 are listed in Table 2.

3. Thermodynamic models

The Gibbs energy function ${}^0G_i^{\varphi}(T) = G_i^{\varphi}(T) - H_i^{SER}$ for the pure elements i (i = C, La, Ce, Pr) in the phase φ is expressed by the following equation [20]:

$$ {}^0G_i^{\varphi}(T) = a + b \cdot T + c \cdot T \cdot \ln T + d \cdot T^2 + e \cdot T^{-1} + f \cdot T^3 + g \cdot T^7 + h \cdot T^{-9} \quad (1) $$

where H_i^{SER} is the molar enthalpy of the element i at 298.15 K and 1 bar in its standard element reference (SER) state. T is the absolute temperature.

The liquid phase is modeled as completely disordered solution phase,and its excess Gibbs energy is described by the Redlich–Kister (R–K) polynomial [53]. The Gibbs energy of the liquid phase is expressed by the following equation:

$$ G^L - H^{SER} = x_C \cdot {}^0G_C^L + x_i \cdot {}^0G_i^L + R \cdot T \cdot [x_C \cdot \ln(x_C) + x_i \cdot \ln(x_i)] + x_C \cdot x_i \cdot \sum_{j=0}^{n} L_{C,i}^{j,L}(x_C - x_i)^j \quad (i = \text{La, Ce, Pr}) \quad (2) $$

where H^{SER} is the abbreviation of $x_C \cdot H_C^{SER} + x_i \cdot H_i^{SER}$, R is the gas constant, and $L_{C,i}^{j,\phi}$ is the jth R–K parameter of liquid phase, which can be expressed as $L_{C,i}^{j,\phi} = a_j + b_j \cdot T$. The coefficients a_j and b_j (j = 0, 1, 2) are the parameters to be optimized.

In view of the reported homogeneity ranges, the solubilities of phases Φ (Φ = Fcc_A1, DHCP, Bcc_A2, La_2C_3, βLaC_2, Pr_2C_3 and βPrC_2) are considered in the present modeling. Since the atomic radiuses of RE (RE = La, Ce, Pr) are much larger than that of C, C must be in the interstitial position of the lattices. Based on the crystal structure of RE (RE = La, Ce, Pr), a two-sublattice model, $(RE)_a(C, Va)_c$ (RE = La, Ce, Pr), is utilized to model the phase Φ. The Gibbs energy of the phase Φ per mole-atoms is expressed by the following equation:

$$ G_{RE:C,Va}^{\Phi} - H^{SER} = y_C'' G_{RE:C}^{\Phi} + y_{Va}'' G_{RE:Va}^{\Phi} + c \cdot RT(y_C'' \ln y_C'' + y_{Va}'' \ln y_{Va}'') + y_C'' y_{Va}'' \left(\sum_{j=0,1,\ldots}^{j} L_{RE:C,Va}^{\Phi}(y_C'' - y_{Va}'')^j \right). \quad (3) $$

Here $G_{RE:C}^{\Phi}$ is the Gibbs energy of the phase Φ in a state in which all interstitial sites are filled with C in the second sublattice. This quantity can be expressed as $G_{RE:C}^{\Phi} - H^{SER} = A + BT + a \cdot {}^0G_{RE}^{ref} + c \cdot {}^0G_C^{graphite}$. $G_{RE:Va}^{\Phi}$ is the Gibbs energy of the phase Φ in a state when the second sublattice is completely occupied by vacant and can be expressed as $G_{RE:Va}^{\Phi} - H^{SER} = {}^0G_{RE}^{\Phi}$. ${}^jL_{RE:C,Va}^{\Phi}$ is the jth parameter of the phase Φ, and can be expressed as ${}^jL_{RE:C,Va}^{\Phi}$ =

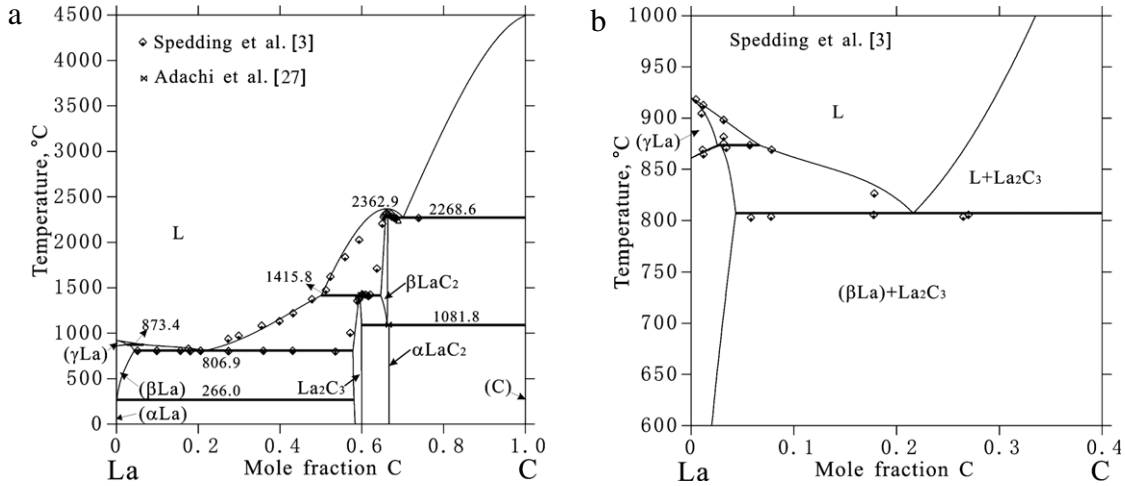

Fig. 1. (a) Calculated C–La phase diagram compared with the experimental data [3,27]; (b) Enlarged La-rich region along with the literature data [3].

$A_j + B_jT$. The interaction parameters A_j and B_j are to be optimized. When the phase Φ stands for Fcc_A1 ((βLa), (γCe)) or DHCP ((αLa), (αPr)), a and c equal to be 1/2; when the phase Φ stands for Bcc_A2 ((γLa), (δCe), (βPr)), a and c equal to be 1/4 and 3/4, respectively; when the phase Φ stands for Pr_2C_3, a and c equal to be 2/5 and 3/5, respectively; and when the phase Φ stands for βLaC_2 or βPrC_2, a and c equal to be 1/3 and 2/3, respectively.

Since there are experimental data on enthalpy increments ($H_T - H_{298\ K}$) of La_2C_3 in a wide temperature range, the Gibbs energy of La_2C_3 per mole-atoms is expressed as:

$$G^{La_2C_3}_{La:C} - 2/5 \cdot H^{SER}_{La} - 3/5 \cdot H^{SER}_{C} = a + b \cdot T + c \cdot T \cdot \ln T + d \cdot T^2 + e \cdot T^{-1}. \quad (4)$$

The coefficients c, d, and e can be optimized from the experimental enthalpy increment data.

In view of negligible homogeneities for αLaC_2, αPrC_2, Ce_2Ce_3, αCeC_2 and βCeC_2 compounds, these phases are treated as stoichiometric ones. The Gibbs energy of each phase per mole-atoms is given by the following expression:

$$G(RE_{(1-x)}C_x) - H^{SER} = A + B \cdot T + (1-x) \cdot {}^0G^{ref}_{Re} + x \cdot {}^0G^{Graphite}_{C} \quad (5)$$

where A and B are the parameters to be evaluated in the course of optimization. $1 - x$ and x are atomic fractions of Re and C for each specific compound. For instance, $1 - x$ and x are equal to be 1/3 and 2/3 for αLaC_2, αPrC_2, αCeC_2 and βCeC_2. ${}^0G^{ref}_{Re}$ denotes the Gibbs energy of pure RE in reference state, as shown in Eq. (1). The reference state for La or Pr is DHCP, while that for Ce is Fcc_A1.

4. Results and discussion

The optimization of the thermodynamic parameters was carried out by using the PARROT module of the Thermo-Calc software, which is based on a least square procedure. The step-by-step optimization procedure carefully described by Du et al. [54] is employed in the present work. The experimental phase diagram data and thermodynamic properties were used as input into the program. Each piece of selected information was given a certain weight based on the uncertainties of the experimental data, and changed by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits.

Considering the fact that certain homogeneity ranges exist for the compounds La_2C_3, βLaC_2, Pr_2C_3 and βPrC_2, the optimization can be divided into two steps. In the first step, all of the compounds are approximated as stoichiometric ones. In the second step, the homogeneity ranges for these compounds are considered. This kind of optimization procedure is demonstrated in the Co–Si system [55].

In the first step, all the intermetallic compounds were treated as stoichiometric phases, and the Gibbs energies for them can be described by Eq. (5). The only exception is La_2C_3, which was described by Eq. (4) due to the existence of enthalpy increment data. We employed the thermodynamic data, including ($H_T - H_{298\ K}$) of La_2C_3 and ΔH_f for all the intermetallic compounds, to assess c, d and e in Eq. (4) for La_2C_3, as well as a and A in Eqs. (4) and (5) for all the compounds. Then, the liquidus and invariant reaction data were included in the assessment to evaluate the thermodynamic parameters of liquid and solid solution phases, as well as b in Eq. (4) and B in Eq. (5) for compounds. Subsequently, all the thermodynamic parameters were optimized simultaneously to describe the general feature of the phase diagram.

In the second step, the homogeneity ranges of the La_2C_3, βLaC_2, Pr_2C_3 and βPrC_2 phases were taken into account. All the thermodynamic parameters obtained in the first step were fixed first. Next, the parameters for other end-members in those compounds were evaluated based on the corresponding experimental data. Finally, all the thermodynamic parameters were optimized simultaneously based on all the reliable experimental data in order to get a self-consistent set of thermodynamic parameters. The thermodynamic parameters finally obtained for the C–RE (RE = La, Ce, Pr) systems are listed in Table 3. The calculated compositions and temperatures for the invariant reactions in comparison with the corresponding experimental data in the C–RE (RE = La, Ce, Pr) systems are listed in Table 4.

The calculated C–La phase diagram, compared with the experimental data, is shown in Fig. 1(a). Fig. 1(b) presents the enlarged C–La phase diagram in the composition range of 0–40 at.% C from 600 to 1000 °C. As can be seen from these figures, the calculated C–La phase diagram agrees well with most of the experimental data except for certain deviation from the liquidus around La-rich side of βLaC_2 phase. One of the possible reasons for the discrepancy is due to the fact that such liquidus temperatures cannot be accurately measured by means of equilibrium-diffusion method. The thermal effects of liquidus should be very weak because the temperature gaps between the invariant ($L + \beta LaC_2 \leftrightarrow La_2C_3$) and liquidus are too large (about 800 °C here). The calculated temperature for (βLa) + La_2C_3 ↔ (αLa) is about 44 °C lower than the experimental one [3], as can be seen in Table 4. This large deviation can be traced to the large differences (i.e.

Table 3
Summary of the thermodynamic parameters in the C–RE (RE = La, Ce, Pr) systems[a].

System	Phase	Sublattice model	Evaluated parameters
C–La	Liquid	$(C, La)_1$	${}^0L^{Liquid}_{C,La} = -259\,436.3 + 30.0 \cdot T$
			${}^1L^{Liquid}_{C,La} = -159\,860.0$
			${}^2L^{Liquid}_{C,La} = -44\,046.6$
	DHCP	$(La)_{1/2}(C, Va)_{1/2}$	${}^0G^{DHCP}_{La:Va} = 1/2 \cdot {}^0G^{DHCP}_{La}$
	Fcc_A1	$(La)_{1/2}(C, Va)_{1/2}$	${}^0G^{Fcc_A1}_{La:Va} = 1/2 \cdot {}^0G^{Fcc_A1}_{La}$
			${}^0G^{Fcc_A1}_{La:C} = -8642.7 + 1/2 \cdot {}^0G^{Fcc_A1}_{La} + 1/2 \cdot {}^0G^{Graphite}_{C}$
	Bcc_A2	$(La)_{1/4}(C, Va)_{3/4}$	${}^0G^{Bcc_A2}_{La:Va} = 1/4 \cdot {}^0G^{Bcc_A2}_{La}$
			${}^0G^{Bcc_A2}_{La:C} = -4091.2 + 1/4 \cdot {}^0G^{Bcc_A2}_{La} + 3/4 \cdot {}^0G^{Graphite}_{C}$
	La_2C_3	$(La)_{2/5}(C, Va)_{3/5}$	${}^0G^{La_2C_3}_{La:C} = -37\,840.1 + 120.6 \cdot T$
			$-20.47 \cdot T \cdot \ln T - 0.00414 \cdot T^2 + 97\,303.5 \cdot T^{-1}$
			${}^0G^{La_2C_3}_{La:Va} = 20\,000 + 2/5 \cdot {}^0G^{DHCP}_{La}$
			${}^0L^{La_2C_3}_{La:C,Va} = -22\,000 + 12.5 \cdot T$
	α LaC_2	$(La)_{1/3}(C)_{2/3}$	${}^0G^{\alpha LaC_2}_{La:C} = -28\,843.2 + 5.0 \cdot T + 1/3 \cdot {}^0G^{DHCP}_{La} + 2/3 \cdot {}^0G^{Graphite}_{C}$
	β LaC_2	$(La)_{1/3}(C, Va)_{2/3}$	${}^0G^{\beta LaC_2}_{La:C} = -5747.6 - 11.8 \cdot T + 1/3 \cdot {}^0G^{DHCP}_{La} + 2/3 \cdot {}^0G^{Graphite}_{C}$
			${}^0G^{\beta LaC_2}_{La:Va} = 20\,000 + 1/3 \cdot {}^0G^{DHCP}_{La}$
			${}^0L^{\beta LaC_2}_{La:C,Va} = -31\,517.6 + 14.1 \cdot T$
C–Ce	Liquid	$(C, Ce)_1$	${}^0L^{Liquid}_{C,Ce} = -268\,285.3 + 22.2 \cdot T$
			${}^1L^{Liquid}_{C,Ce} = -139\,372.9$
			${}^2L^{Liquid}_{C,Ce} = -24\,515.3$
	Fcc_A1	$(Ce)_{1/2}(C, Va)_{1/2}$	${}^0G^{Fcc_A1}_{Ce:Va} = 1/2 \cdot {}^0G^{Fcc_A1}_{Ce}$
			${}^0G^{Fcc_A1}_{Ce:C} = -16\,978.0 + 1/2 \cdot {}^0G^{Fcc_A1}_{Ce} + 1/2 \cdot {}^0G^{Graphite}_{C}$
	Bcc_A2	$(Ce)_{1/4}(C, Va)_{3/4}$	${}^0G^{Bcc_A2}_{Ce:Va} = 1/4 \cdot {}^0G^{Bcc_A2}_{Ce}$
			${}^0G^{Bcc_A2}_{Ce:C} = -16\,163.7 + 1/4 \cdot {}^0G^{Fcc_A1}_{Ce} + 3/4 \cdot {}^0G^{Graphite}_{C}$
	Ce_2C_3	$(Ce)_{2/5}(C)_{3/5}$	${}^0G^{Ce_2C_3}_{Ce:C} = -35\,851.3 + 1.75 \cdot T + 2/5 \cdot {}^0G^{Fcc_A1}_{Ce} + 3/5 \cdot {}^0G^{Graphite}_{C}$
	α CeC_2	$(Ce)_{1/3}(C)_{2/3}$	${}^0G^{\alpha CeC_2}_{Ce:C} = -32\,398.3 + 3.29 \cdot T + 1/3 \cdot {}^0G^{Fcc_A1}_{Ce} + 2/3 \cdot {}^0G^{Graphite}_{C}$
	β CeC_2	$(Ce)_{1/3}(C)_{2/3}$	${}^0G^{\beta CeC_2}_{Ce:C} = -14\,973.2 - 9.39 \cdot T + 1/3 \cdot {}^0G^{Fcc_A1}_{Ce} + 2/3 \cdot {}^0G^{Graphite}_{C}$
C–Pr	Liquid	$(C, Pr)_1$	${}^0L^{Liquid}_{C,Pr} = -206\,169.6 - 29.9 \cdot T$
			${}^1L^{Liquid}_{C,Pr} = -127\,621.8$
			${}^2L^{Liquid}_{C,Pr} = -26\,606.5$
	DHCP	$(Pr)_{1/2}(C, Va)_{1/2}$	${}^0G^{DHCP}_{Pr:Va} = 1/2 \cdot {}^0G^{DHCP}_{Pr}$
			${}^0G^{DHCP}_{Pr:C} = -18\,424.2 + 1/2 \cdot {}^0G^{DHCP}_{Pr} + 1/2 \cdot {}^0G^{Graphite}_{C}$
	Bcc_A2	$(Pr)_{1/4}(C, Va)_{3/4}$	${}^0G^{Bcc_A2}_{Pr:Va} = 1/4 \cdot {}^0G^{Bcc_A2}_{Pr}$
			${}^0G^{Bcc_A2}_{Pr:C} = 55\,543.8 + 1/4 \cdot {}^0G^{Bcc_A2}_{Pr} + 3/4 \cdot {}^0G^{Graphite}_{C}$
			${}^0L^{Bcc_A2}_{Pr:C,Va} = -82\,918.2$
	Pr_2C_3	$(Pr)_{2/5}(C, Va)_{3/5}$	${}^0G^{Pr_2C_3}_{Pr:C} = -29\,469.3 - 5.7 \cdot T$
			$+2/5 \cdot {}^0G^{DHCP}_{Pr} + 3/5 \cdot {}^0G^{Graphite}_{C}$
			${}^0G^{Pr_2C_3}_{Pr:Va} = 20\,000 + 2/5 \cdot {}^0G^{DHCP}_{Pr}$
			${}^0L^{Pr_2C_3}_{Pr:C,Va} = -18\,844.6$
	α PrC_2	$(Pr)_{1/3}(C)_{2/3}$	${}^0G^{\alpha PrC_2}_{Pr:C} = -28\,186.5 - 2.2 \cdot T + 1/3 \cdot {}^0G^{DHCP}_{Pr} + 2/3 \cdot {}^0G^{Graphite}_{C}$
	β PrC_2	$(Pr)_{1/3}(C, Va)_{2/3}$	${}^0G^{\beta PrC_2}_{Pr:C} = -2509.9 - 20.2 \cdot T + 1/3 \cdot {}^0G^{DHCP}_{Pr} + 2/3 \cdot {}^0G^{Graphite}_{C}$
			${}^0G^{\beta PrC_2}_{Pr:Va} = 20\,000 + 1/3 \cdot {}^0G^{DHCP}_{Pr}$
			${}^0L^{\beta PrC_2}_{Pr:C,Va} = -3776.0$

[a] Gibbs energy in J/mol-atom, temperature (T) in Kelvin and pressure (P) in bar. The Gibbs energies for the pure elements are from the SGTE compilation [20].

Table 4
Calculated invariant equilibria in the C–RE (RE = La, Ce, Pr) systems compared with the literature data.

Systems	Invariant reactions	Composition (at.% C)			T, °C	Method	Source
C–La	$(\beta La) + La_2C_3 \leftrightarrow (\alpha La)$	0	58.1	0	266.0	Calculated	Present work
		0[a]	∼55.8	0[a]	310 ± 5	Dilatometric	Measured [3]
	$L \leftrightarrow (\beta La) + La_2C_3$	20.9	4.5	57.8	806.9	Calculated	Present work
		20.6	∼4.4[a]	∼54.8[a]	806 ± 2	PA/TA	Measured [3]
	$L + (\gamma La) \leftrightarrow (\beta La)$	6.3	2.5	2.8	873.4	Calculated	Present work
		∼5.7[a]	∼2.8[a]	∼3.8[a]	875 ± 1	TA	Measured [3]
	$\beta LaC_2 \leftrightarrow LaC_2C_3 + \alpha LaC_2$	66.2	59.9	66.7	1081.8	Calculated	Present work
		?	?	?	1081 ± 2	DTA	Measured [27]
	$\beta LaC_2 + (C) \leftrightarrow \alpha LaC_2$	66.2	100.0	66.7	1081.8	Calculated	Present work
		?	?	?	1081 ± 2	DTA	Measured [27]
	$L + \beta LaC_2 \leftrightarrow La_2C_3$	50.0	64.5	59.3	1415.8	Calculated	Present work
		∼50.6[a]	∼63.1[a]	∼60.0[a]	1415 ± 3	PA	Measured [3]
	$L \leftrightarrow \beta LaC_2 + (C)$	70.1	66.3	100.0	2268.6	Calculated	Present work
		?	∼67.0[a]	?	2271 ± 15	PA	Measured [3]
	$L \leftrightarrow \beta LaC_2$	66.0	66.0		2362.9	Calculated	Present work
		∼66.1[a]	∼66.1[a]		2356 ± 25	Extrapolation[b]	Measured [3]
C–Ce	$L \leftrightarrow (\gamma Ce) + Ce_2C_3$	19.2	4.4	60.00	638.3	Calculated	Present work
	$L + (\delta Ce) \leftrightarrow (\gamma Ce)$	6.2	1.3	1.9	744.0	Calculated	Present work
	$\beta CeC_2 + (C) \leftrightarrow \alpha CeC_2$	66.7	100.0	66.7	1099.8	Calculated	Present work
		?	?	?	1100 ± 20	KCMS	Measured [15]
	$\beta CeC_2 \leftrightarrow Ce_2C_3 + \alpha CeC_2$	66.7	60.00	66.7	1099.8	Calculated	Present work
		?	?	?	1100 ± 20	KCMS	Measured [15]
	$L + \beta CeC_2 \leftrightarrow Ce_2C_3$	50.5	66.67	60.00	1529.9	Calculated	Present work
		?	66.67	60.00	1530 ± 30	TA	Measured [23,47]
	$L \leftrightarrow \beta CeC_2 + (C)$	68.9	66.67	100.0	2269.1	Calculated	Present work
		?	66.67	100.0	2270 ± 20	KCMS	Measured [15]
	$L \leftrightarrow \beta CeC_2$	66.67	66.67		2290.3	Calculated	Present work
		66.67	66.67		2290	TA	Measured [23, 48–50]
C–Pr	$(\beta Pr) \leftrightarrow Pr_2C_3 + (\alpha Pr)$	6.5	54.8	4.4	675.9	Calculated	Present work
		∼6.2[a]	?	?	675 ± 6	DTA	Measured [5]
	$L \leftrightarrow (\beta Pr) + Pr_2C_3$	15.5	7.7	54.4	797.9	Calculated	Present work
		∼15	?	?	800 ± 4	DTA	Measured [5]
	$\beta PrC_2 \leftrightarrow Pr_2C_3 + \alpha PrC_2$	66.6	59.8	66.7	1151.4	Calculated	Present work
		?	?	?	1145 ± 4	DTA	Measured [5]
	$\beta PrC_2 + (C) \leftrightarrow \alpha PrC_2$	66.6	100.0	66.67	1151.4	Calculated	Present work
		?	?	?	1134 ± 4	DTA	Measured [5]
	$L + \beta PrC_2 \leftrightarrow Pr_2C_3$	50.0	65.7	58.1	1545.6	Calculated	Present work
		50	?	?	1545 ± 4	DTA	Measured [5]
	$L \leftrightarrow \beta PrC_2 + (C)$	70.4	66.5	100.0	2248.5	Calculated	Present work
		?	?	100.0	2254 ± 6	DTA	Measured [5]
	$L \leftrightarrow \beta PrC_2$	66.4	66.4		2320.1	Calculated	Present work
		?	?		2320	Extrapolation[b]	Measured [5]

TA = Thermal analysis.
DTA = Differential thermal analysis.
PA = Pirani and Alterthums method.
KCMS = Knudsen cell mass spectrometer.
[a] Indicates that the values are read from the original graphs.
[b] Indicates that the melting temperatures are extrapolated by the observed melting points of alloys near the theoretical composition.

33 °C) between the experimental transformation temperature of $(\alpha La) \leftrightarrow (\beta La)$ by Spedding et al. [3] and the thermodynamic data of pure La for calculation (i.e. 277 °C) according to Dinsdale [20]. In addition, the calculated melting temperature for LaC_2 is 6.9 °C higher than the experimental value, which is extrapolated by the melting points near the theoretical composition and has a big deviation for 25 °C [3]. In view of this, the calculated result is regarded to be acceptable.

The calculated enthalpies of formation in the C–La system are compared with the experimental data, as presented in Fig. 2. Considering the fact that the experimental data show some scattering for LaC_2, the present result agrees best with that [35]. It can be seen that the calculated enthalpy of formation for La_2C_3 is much smaller than the experimental value [40]. The dashed line indicates a situation that the value for La_2C_3 fits well with the experimental data while that for LaC_2 does not. It is worthwhile to notice that Anderson and Bagshaw citation [40] measured the enthalpy of formation for La_2C_3 and two years later they determined another composition (LaC_2) by using the same method citation [35]. In the present work, the latest experiment is considered to be more reliable, and thus a calculation result as

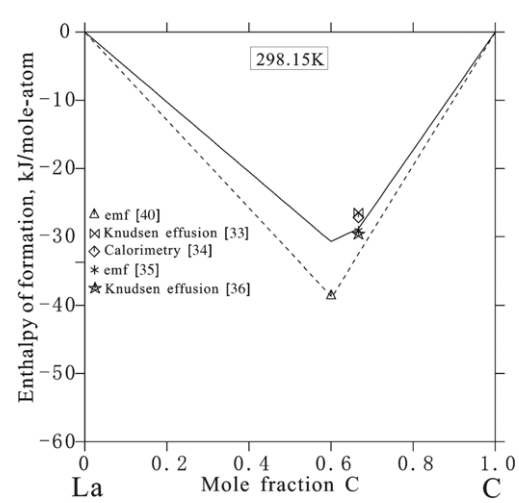

Fig. 2. Calculated enthalpies of formation at 298.15 K in the C–La system, compared with the experimental data [33–36,40].

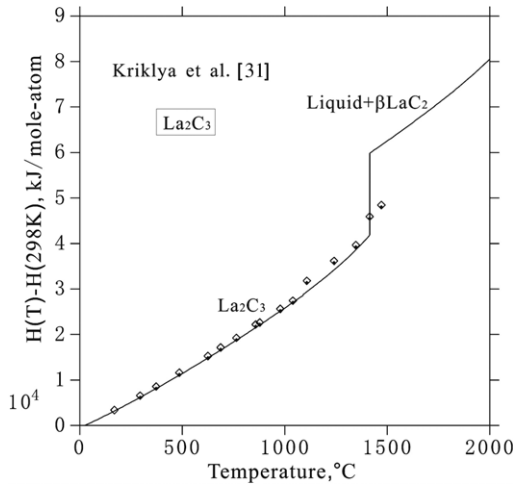

Fig. 3. Calculated $H_T - H_{298\ K}$ for the compound La_2C_3 along with the measured data [31].

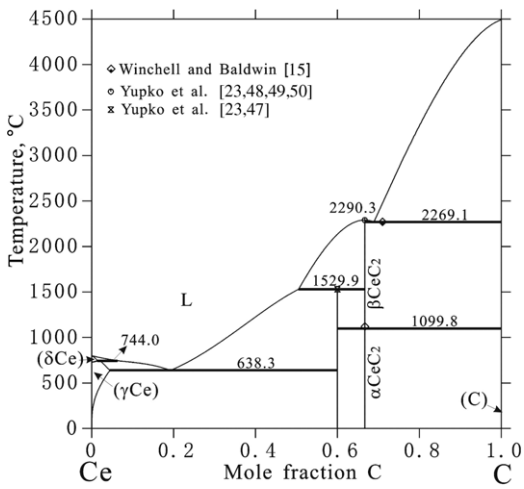

Fig. 4. Calculated C–Ce phase diagram along with the experimental data [23,47,48,15,49,50].

the solid line is conducted. The calculated enthalpy increment for the compound La_2C_3 compared with the experimental data [31] is shown in Fig. 3. The agreement is good within the estimated experimental uncertainties, except for deviations at temperatures close to the melting point, where large experimental errors are expected at such high temperatures.

Fig. 4 presents the calculated C–Ce phase diagram using the presently obtained parameters. It can be seen that the present calculation shows an excellent agreement with the selected experimental data. The calculated enthalpies of formation for the C–Ce system compared with the experimental data are presented in Fig. 5. Similarly, the calculated enthalpy of formation for Ce_2C_3 and CeC_2 are consistent with the experimental data [41].

The calculated C–Pr phase diagram compared with the experimental data is presented in Fig. 6. It is seen that the calculated results show a reasonable agreement with the experimental information. The calculated enthalpies of formation at 298.15 K for the C–Pr system compared with the experimental data are presented in Fig. 7. As can be seen, the presently calculated results agree well with all the experimental data [35,40,43].

Due to the similar properties of the rare earth elements La, Ce, and Pr, the C–La, C–Ce and C–Pr phase diagrams possess the similar topology phase relation, as also indicated in the present work. Moreover, the numerical ranges for the thermodynamic

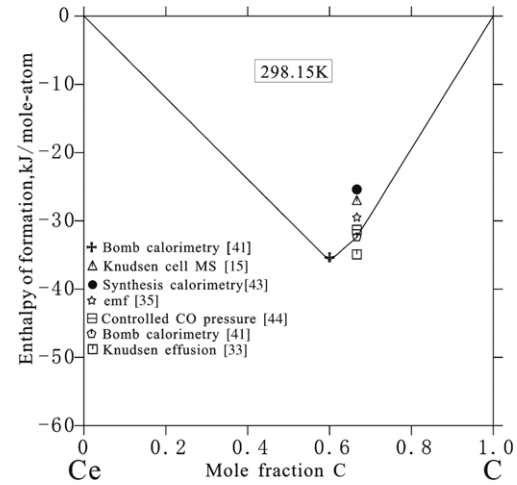

Fig. 5. Calculated enthalpies of formation for the C–Ce system at 298.15 K, compared with the experimental data [33,35,15,44,41,43].

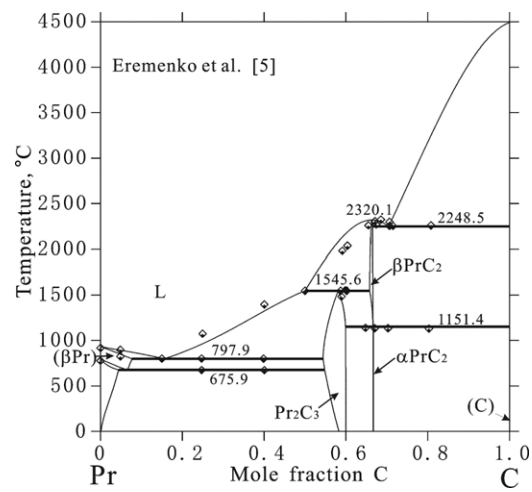

Fig. 6. Calculated C–Pr phase diagram compared with the experimental data [5].

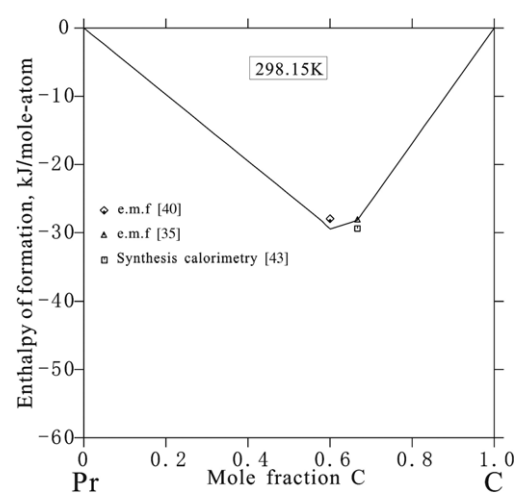

Fig. 7. Calculated enthalpies of formation for the C–Pr system at 298.15 K, compared with the experimental data [35,40,43].

parameters in liquid phase are more or less the same for the three binary systems. Due to the large asymmetry of the liquidus, the parameters a_1 in Eq. (2) present huge values. The entropies for the intermetallic compounds (RE_2C_3 and REC_2) at 298.15 K for the C–RE (RE = La, Ce and Pr) binary systems are presented

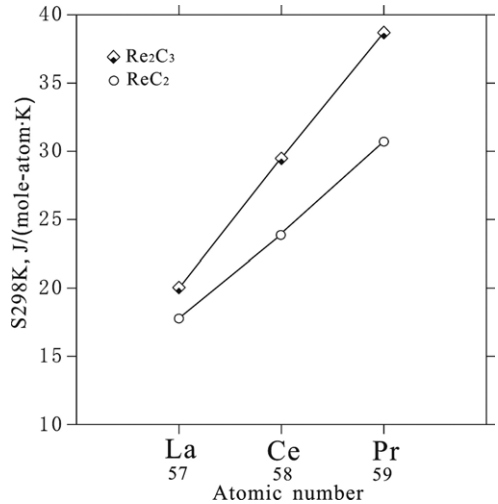

Fig. 8. The calculated entropies of the intermetallic compounds (RE_2C_3 and REC_2) at 298 K for the C–RE (RE = La, Ce and Pr) binary systems.

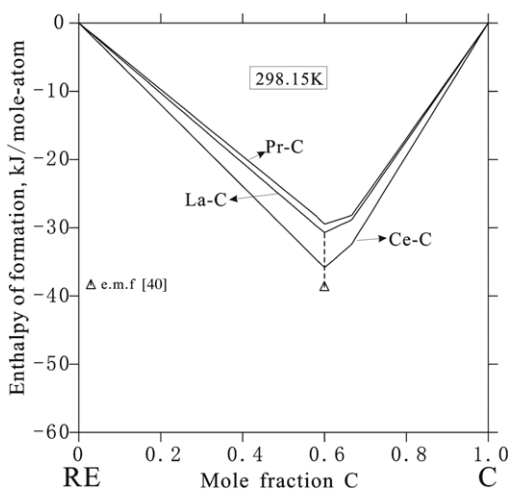

Fig. 9. Calculated enthalpies of formation for the three binary systems (C–La, C–Ce, and C–Pr) according to the presently obtained thermodynamic parameters.

in Fig. 8, from which we can see that the entropy for RE_2C_3 or REC_2 at 298.15 K increases with the increase of atomic number. This information can be helpful for the investigation on the phase diagram and thermodynamic properties between *C* and other rare earth elements, especially for those with limited experimental information. In addition, we plot the calculated enthalpies of formation at 298.15 K in these three systems for a direct comparison, as presented in Fig. 9. It can be clearly seen that the curves are very similar. The calculated enthalpies of formation at 298.15 K for these three binary systems agree well with the experimental data, as can been seen in Figs. 2, 5 and 7, except for the curve of the C–La system at the composition of 60 at.% C. This indicates that further measurement of La_2C_3 compound is highly needed.

5. Conclusions

- All the phase diagram data and thermodynamic properties of the C–RE (RE = La, Ce and Pr) binary systems available in the literature were critically eviewed.
- Based on the critically reviewed literature data, self-consistent thermodynamic parameters were finally obtained for each binary system by using the CALPHAD approach. The calculated phase diagrams and thermodynamic properties agree reasonably with most of the literature data.
- Thermodynamic similarities among these three systems were demonstrated by comparing the thermodynamic parameters of liquid phase and the entropies of the intermetallic compounds at 298.15 K in these three systems. Such similarities are of some guidance for studying other carbon–rare earth systems.

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Appendix. Supplementary data

Supplementary material related to this article can be found online at doi:10.1016/j.calphad.2011.09.002.

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